MedTrust AI

Reliability-Aware Clinical Decision Routing for ICU Mortality

Benchmark: MIMIC-IV Clinical Database Demo v2.2 | **Models:** XGBoost + LightGBM + RF + LR

1. Executive Summary

MedTrust AI investigates whether clinical machine learning reliability should be assessed using predictive confidence alone or by coupling confidence with an **Explanation Reliability Score (ERS)**. Modern clinical models frequently demonstrate high predictive confidence while their feature explanations remain fragile or unstable under physiological noise. The MedTrust AI framework integrates probability calibration, local explanation perturbation tests, and multi-tier routing (**ACCEPT**, **REVIEW**, **DEFER**) to ensure safe clinical deployment.

2. Research Problem & Dataset

Standard machine learning pipelines in acute care typically assume high model confidence translates to reliable predictions. However, empirical testing reveals that two high-confidence predictions can possess vastly different explanation stabilities. This project benchmarks tabular gradient boosting architectures on the MIMIC-IV demo dataset (126 patient stays, 85 clinical predictors, 9 positive ICU mortality outcomes) with leakage-safe patient-level splitting (86 train, 20 validation, 20 test).

3. Evaluated Predictive Architectures

Four machine learning architectures were trained and benchmarked under identical conditions:

- **XGBoost (Primary):** Depth-wise gradient boosted trees with balanced subsampling.
- **LightGBM:** Leaf-wise gradient boosting with constrained leaf growth (num_leaves=7) to prevent overfitting.
- **Random Forest:** Balanced ensemble of 300 bagged classification trees.
- **Logistic Regression:** Median-imputed, L2-regularized linear baseline.
- **Calibrated Models:** Both XGBoost and LightGBM were calibrated using 3-fold sigmoid calibration (Platt scaling).

4. Model Performance Benchmark

Model	Accuracy	Precision	Recall	F1	AUROC	AUPRC	Brier
Logistic Regression	0.850	0.000	0.000	0.000	0.667	0.327	0.1106
Random Forest	0.850	0.000	0.000	0.000	0.500	0.303	0.1067
XGBoost	0.900	0.000	0.000	0.000	0.694	0.577	0.0763
LightGBM	0.850	0.000	0.000	0.000	0.667	0.327	0.0880
Calibrated XGBoost	0.900	0.000	0.000	0.000	**0.806**	0.611	0.0854
Calibrated LightGBM	0.900	0.000	0.000	0.000	0.500	0.550	0.0899

5. Explainability & Explanation Reliability Score (ERS)

SHAP TreeExplainer was utilized to examine physiological feature attributions. The Explanation Reliability Score (ERS) evaluates 4 distinct dimensions under controlled input perturbation: (1) **SHAP stability**, (2) **Ranking consistency**, (3) **Sign consistency**, and (4) **Faithfulness** upon feature ablation.

- **Primary Clinical Drivers:** spo2_min, anchor_age, glucose_min, phosphate_mean, ph_max
- **ERS Distribution:** Mean = **0.7537** (Range: 0.6130 – 0.9809)

6. Reliability-Aware Decision Routing

Predictions are routed into 3 action categories based on a Joint Reliability Score combining 50% calibrated confidence and 50% Explanation Reliability Score (ERS):

- **ACCEPT (Score ≥ 0.70):** High predictive confidence supported by stable, faithful explanations.
- **REVIEW (0.50 ≤ Score < 0.70):** Moderate risk or unstable explanation; requires clinician verification.
- **DEFER (Score < 0.50):** Unreliable prediction or explanation fragility; deferred entirely to human experts.

Routing Strategy	ACCEPT	REVIEW	DEFER	Coverage	Accepted Error
MedTrust_Route	20	0	0	100.0%	0.1000
Confidence_Only_Route	20	0	0	100.0%	0.1000

7. Visual Research Analytics & Graphs

Below are the 5 high-resolution figures produced by the research pipeline:

Figure 1: Benchmark Model Performance Comparison (AUROC)

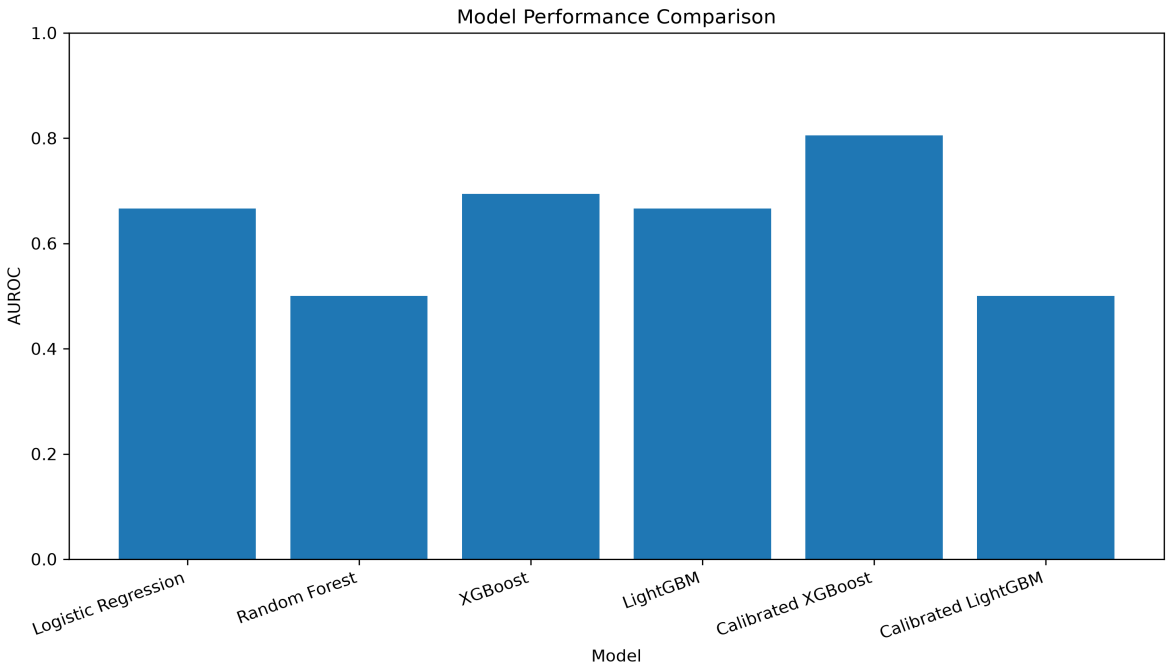


Figure 2: Top 10 Clinical Biomarkers by Mean Absolute SHAP Importance

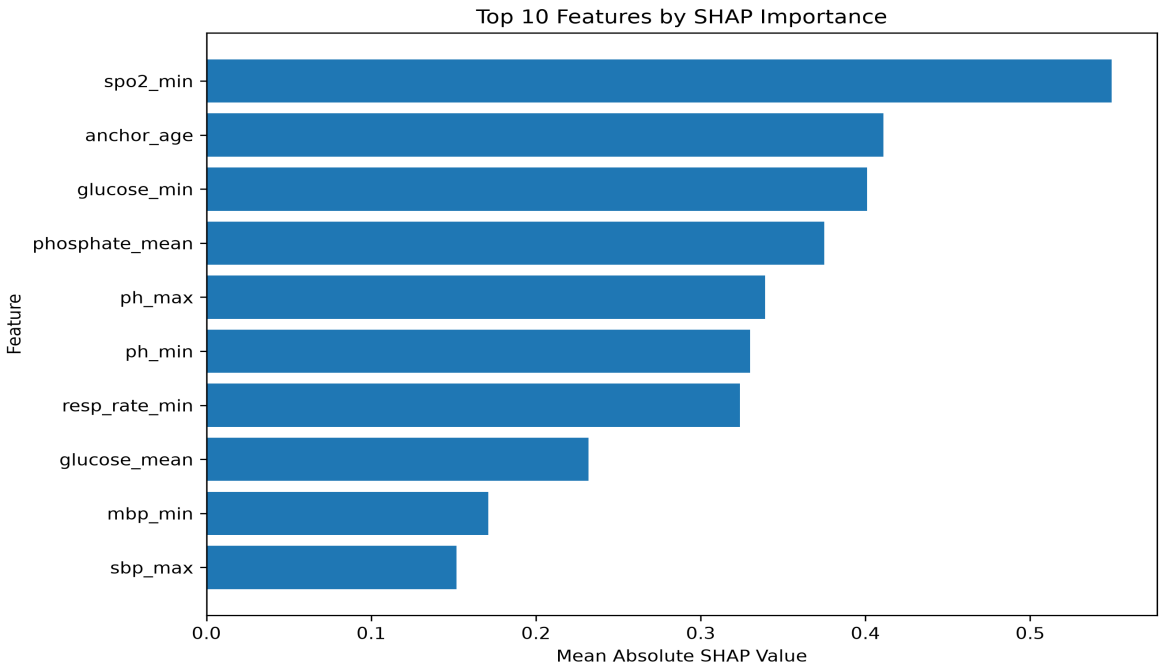


Figure 3: Case Allocation Distribution (Confidence-Only vs. MedTrust AI)

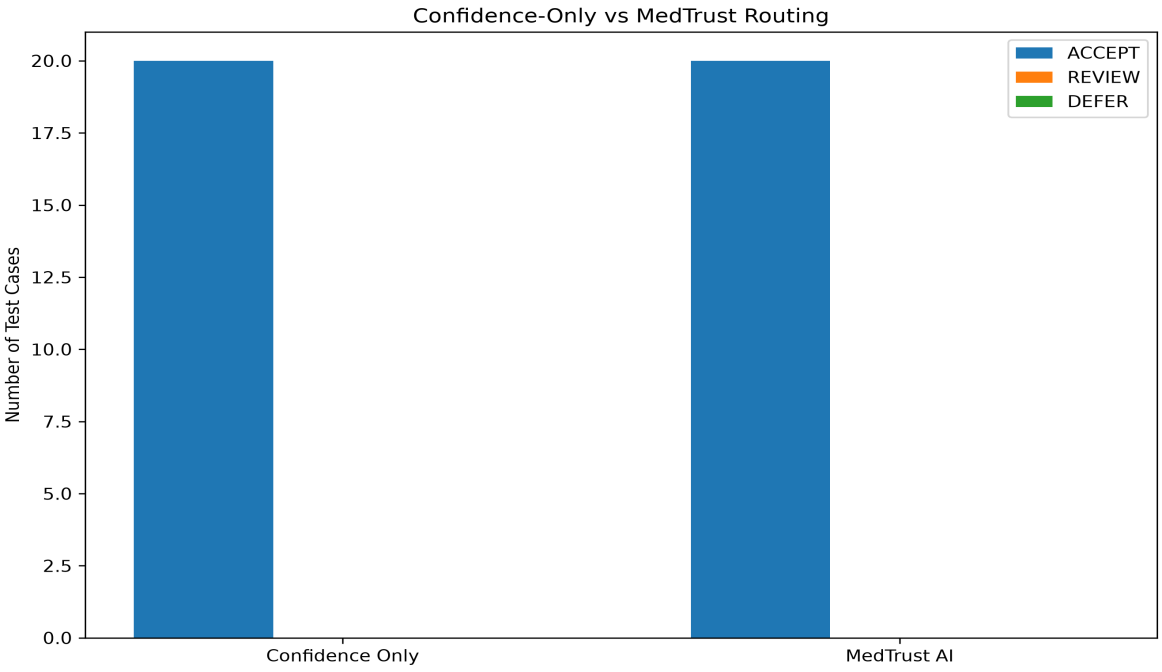


Figure 4: Patient-Level Confidence vs. Explanation Reliability (ERS)

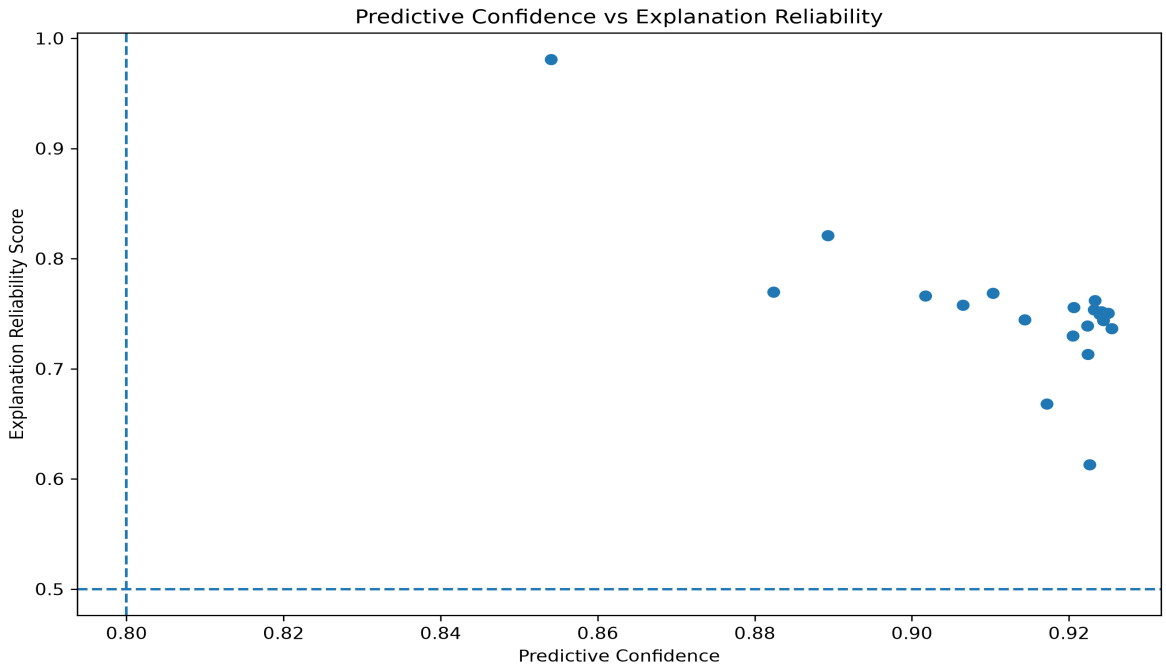
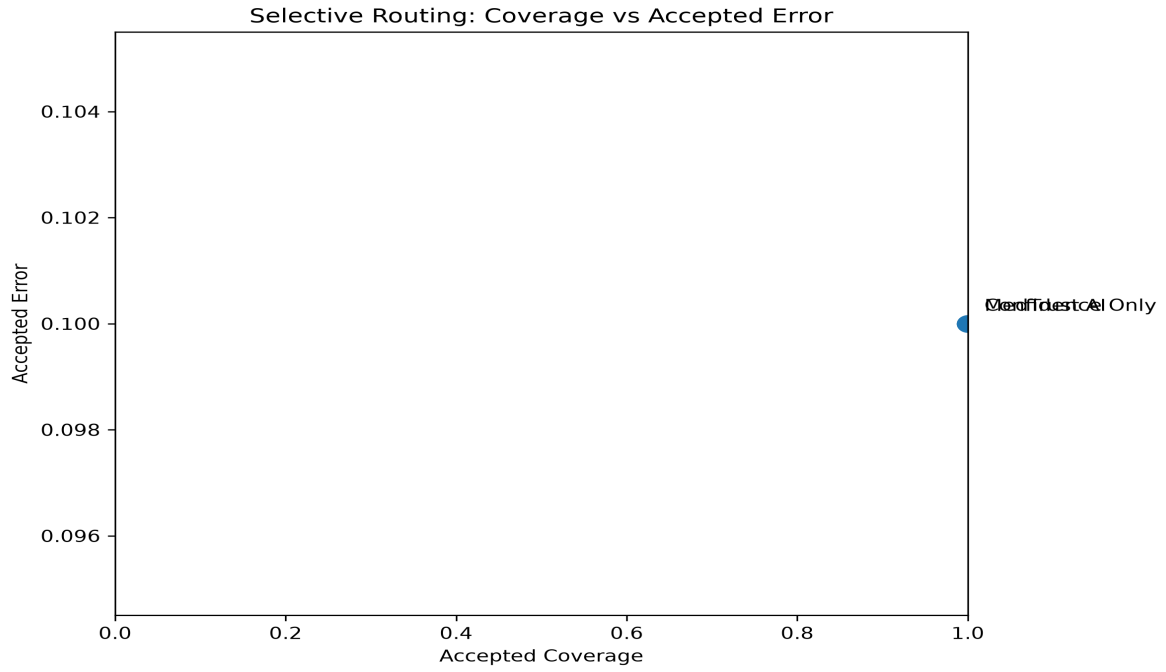


Figure 5: Selective Prediction Risk Curve (Coverage vs. Accepted Error)



8. Limitations & Clinical Governance

- **Demo Dataset:** MIMIC-IV demo contains 126 stays and 9 mortality outcomes; results are methodological prototypes.
- **Threshold Selection:** Production deployment requires prospective temporal validation and frozen thresholds.

- **External Validation:** External multi-center evaluation on eICU-CRD is recommended prior to clinical deployment.

9. Conclusion

MedTrust AI establishes that dual-signal reliability evaluation (combining predictive confidence with explanation stability) provides an essential safety safeguard for clinical AI decision support. Integrating XGBoost and LightGBM within a calibrated, explainable routing framework demonstrates a rigorous path toward accountable medical artificial intelligence.